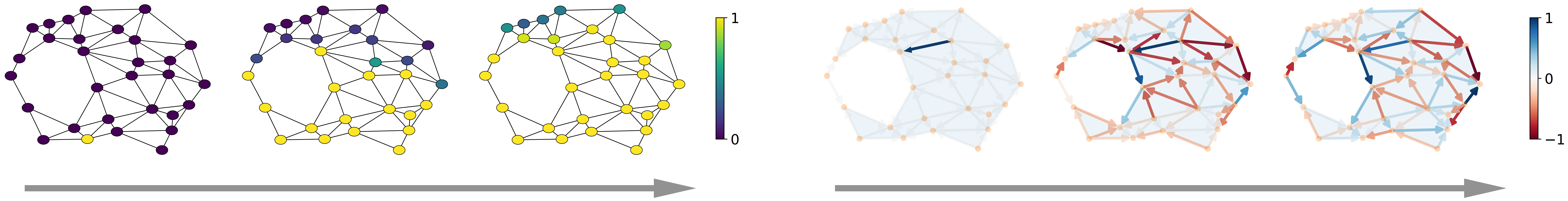
Reference topological dynamics

- $Y \sim \mathbb{Q}_{\mathcal{T}}$: topology-aware stochastic dynamics, tractable

topology-aware
transition kernel ✅

- Topological stochastic heat diffusion: $dY_t = -cLY_t dt + g_t dW_t$



- Topological stochastic dynamics: $dY_t = f(t, Y_t; L)dt + g_t dW_t$
  - $f_t = H_t(L)Y_t + \alpha_t$ with $H_t(L)$ a topological convolution wrt Laplacian L

Stochastic optimal control

- Optimal TSB: a forward-backward SDE system

Enables learning!!!

Forward: $dX_t = dY_t + Z_t dt,\ X_0 \sim \rho_0$

Backward: $dX_t = dY_t - \tilde{Z}_t dt,\ X_1 \sim \rho_1$

Reference:
$dY_t = -cLY_t dt + g_t dW_t$

TSB-learning model

$Z_t \approx Z_t(\theta)$	$l(x_0; \phi)$
$\tilde{Z}_t \approx \tilde{Z}_t(\phi)$	$l(x_1; \theta)$
Learnable	Trainable

- Learnable models $(Z_t(\theta), \hat{Z}_t(\hat{\theta}))$ for optimal policies $(Z_t, \hat{Z}_t)$
  - NNs, GNNs, simplicial NNs
- Trainable objective: max likelihood

$$\mathscr{L}_{TSB}(x_0) = \mathbb{E}\left[\log \nu_1(X_1)\right] - \int_0^1 \mathbb{E}\left[\frac{1}{2}\|Z_t\|^2 + \frac{1}{2}\|\hat{Z}_t\|^2 + \nabla \cdot (g_t \hat{Z}_t - f_t) + \hat{Z}_t^\top Z_t\right] dt$$

- Unifies diffusion model

Diffusion model

$Z_t = 0, \quad \hat{Z}_t = g_t \nabla \log p_{t|0}$